# 50m / 150m / 200m Industrial High Frequency

# Laser Rangefinder Module



## Key Properties

- Measurement Range: 0.05m ~ 50/150/200m
- Measurement Frequency: 20~20KHz
- Measurement Accuracy: <100m: ±0.05m
  ≥100m: ±0.1m
- Wavelength: 905 nm
- Laser Safety: Class1 / Class3
-Communication Interface: TTL

## Function

- Continuous Measurement
- Frequency Setting
- Baudrate Setting

## Introduction

LRFX0M20KHS is a high-frequency rangefinder module designed to deliver exceptional performance. Its maximum frequency can reach 20KHz, and users can configure the frequency from 20 to 20KHz through commands. With its compact size and lightweight design, it's perfect for seamless integration into various devices.

The LRFX0M20KHS is suitable for both indoor and outdoor use, boasting measurement accuracy down to centimeter levels. We offer three measurement distance options: 50 meters, 150 meters, and 200 meters, catering to diverse needs. Experience precision, versatility, and convenience all in one advanced solution.

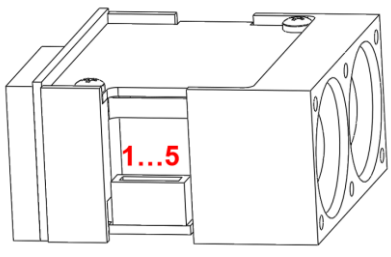
## 1. Technical specifications

| Technical specifications | | | |
|---|---|---|---|
| Model Name | LRF50M20KHS | LRF150M20KHS | LRF200M20KHS |
| Max Range | 50m | 150m | 200m |
| Min Range | 0.05m | | |
| Frequency | 20~20KHz | | |
| Accuracy | <100m: ±0.05m<br>≥100m: ±0.1m | | |
| **Optical Design** | | | |
| Wavelength | 905nm | | |
| Laser Safety | <2500Hz: Class1<br>≥2500Hz: Class3 | <250Hz: Class1<br>≥250Hz: Class3 | <220Hz: Class1<br>≥220Hz: Class3 |
| Divergence | 3 mrad | 6 mrad | |
| Anti-ambient Light | ≥100K LUX | | |
| **Communication** | | | |
| Communication Interface | 3.3V TTL (UART) | | |
| Baudrate | 9600~921600 bps，default as 460800 bps | | |
| Interface Connector | JST-SH-1.25mm-5P | | |
| **Power Consumption** | | | |
| Working voltage | 24V (9~36V) | | |
| Power Consumption | <1W | | |
| **Mechanical** | | | |
| Dimensions | 33 x 36 x 18 mm | | |
| Weight | 20g±1g | | |
| **Environmental** | | | |
| Operating Temperature | -40°C~85°C | | |
| Storage Temperature | -45°C~100°C | | |

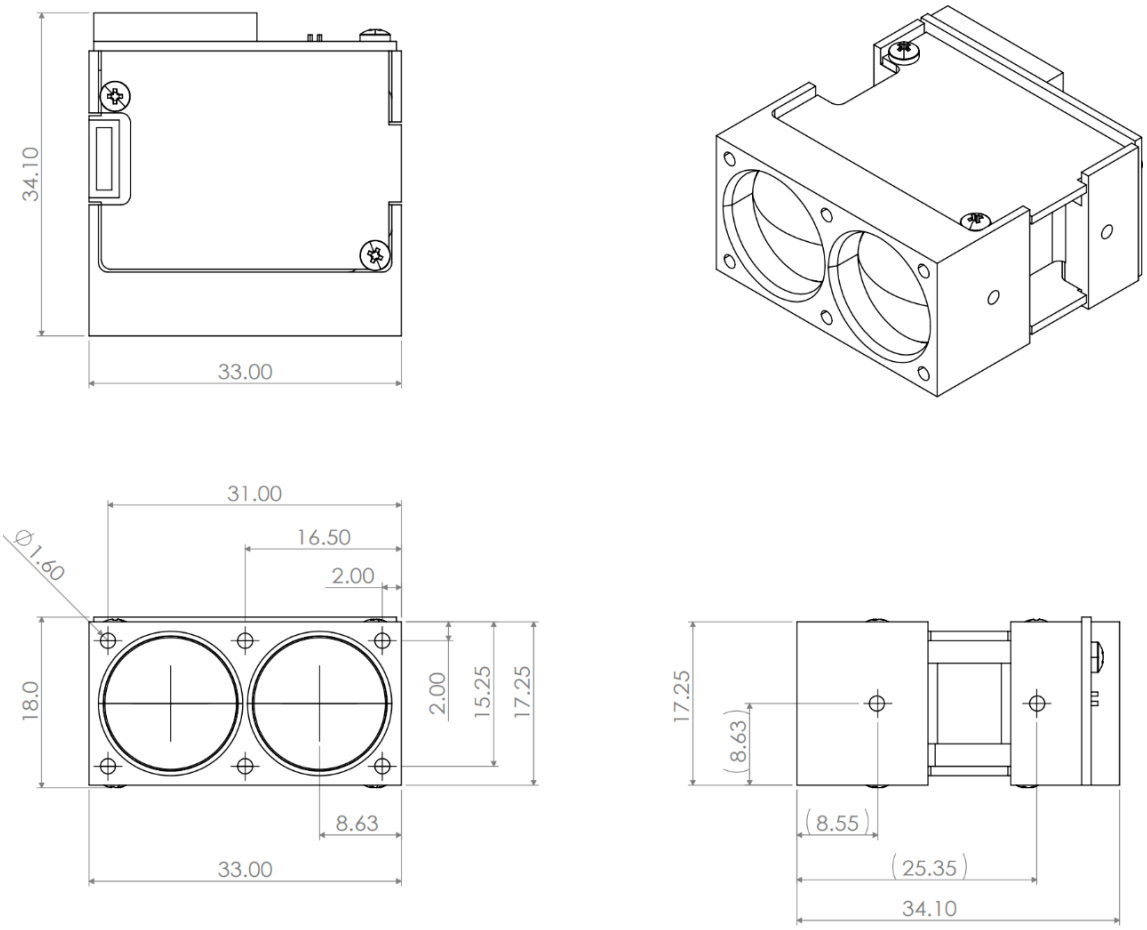
*note: Measurement Range Condition: 85% reflectivity.

## 2. Pin Assignment

| Pin | Color | Description |
|---|---|---|
| 1 | Red | DC 9~36V |
| 2 | Black | Power GND |
| 3 | Yellow | TTL GND |
| 4 | Green | TX (TTL: Voltage is programmable: 3.3V) |
| 5 | Blue | RX (TTL: Voltage is programmable: 3.3V) |



## 3. Dimensions




## 4. Communication

### 4.1 UART configuration parameters:

Baudrate: 9600 / 19200 / 115200 / 256000 / 460800 / 921600 (default as 460800)

Data bits: 8

Parity bits: None

Stop bits: 1

### 4.2 Communication frame format:

After powering on, the ranging module will actively output data (8 bytes per frame).

| Head (1 byte) | Distance (2 byte) | | Checksum (1 byte) |
|---|---|---|---|
| 5C | * | * | * |

*Head: Fixed as 0x5C.

*Distance: Measure result, range: 0~65536, unit: cm.

*Checksum: Function + Length + Distance + Signal Quality and take inverse

Example: 5C 02 11 EC

*5C: Head (Fixed).

*02 11: Distance is 4354cm. (The first byte is the low bit and the second byte is the high bit.)

0x02 + 0x11 * 256= 4354.

*EC: Checksum, 0x02 + 0x11 = 2 + 17 = 19

19 = 0x13

The two's complement of 0x13 is EC.

## 4.3 Communication command

### 4.3.1 Set Frequency

| | |
|---|---|
| Send | 5A 0B 02 **XX XX** CHK |
| Receive | 5A 8B 02 **XX XX** CHK |
| Example | Set frequency as 1000Hz<br>Send: 5A 0B 02 E7 03 08<br>Receive: 5A 8B 02 E7 03 88 |

* The E7 03 in the example represents the setting of the calculation parameter as 999 (in hexadecimal, where E7 is the low byte and 03 is the high byte)

The frequency is set to 1000000/（999+1）=1000Hz.

* The parameter range is from 49999 to 49, corresponding to a frequency range of 20 to 20000Hz.

### 4.3.2 Get Frequency

| | |
|---|---|
| Send | 5A 1B 02 1B 1B AC |
| Receive | 5A 9B 02 **XX XX** CHK |
| Example | Send: 5A 1B 02 1B 1B AC<br>Receive: 5A 9B 02 E7 03 78 |

* The E7 03 in the example represents the setting of the calculation parameter as 999

### 4.3.3 Set Baudrate

| | |
|---|---|
| Send | 5A 06 02 **XX XX** CHK |
| Receive | 5A 86 02 **XX XX** CHK |
| Example | Set baudrate as 115200<br>Send: 5A 06 02 80 04 73<br>Receive: 5A 86 02 80 04 F3 |

*The baudrate table corresponding to the configuration commands is as follows:

| Command | Baudrate |
|---|---|
| 60 00 | 9600 |
| C0 00 | 19200 |
| 80 04 | 115200 |
| 00 0A | 256000 |
| 00 12 | 460800 |
| 00 24 | 921600 |

*When the frequency is <200Hz, any baud rate can be used.

* When frequency >200Hz, the baudrate needs to be set to at least 19200.

* When frequency >400Hz, the baudrate needs to be set to at least 115200.

* When frequency >2500Hz, the baudrate needs to be set to at least 256000.

* When frequency >6000Hz, the baudrate needs to be set to at least 460800.

* When frequency >10500Hz, the baudrate needs to be set to at least 921600.